

Analysing Enhanced EEG Based Brain Computer Interface for Motor Imagery Tasks Using Statistical Analysis

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Abstract—BCIs have emerged as a useful tool for helping people with neurological disorders like epilepsy, ALS, and cerebral palsy, who have severely limited communication, by analyzing EEG signals and turning them into actionable signals. This paper aims to investigate the application of BCIs in analyzing EEG signals and designing them to give meaningful signals to the users. Five healthy, right-handed university students (18–21 years) were selected for the study and were asked to spell the words mentally without vocalizing for four cognitive tasks; forward, stop, left, and right. At 100 Hz, EEG signals were sampled and pre-processed with a notch filter and feature extraction was done using DWT to extract important features and KNN was used for classification. All the subjects showed high accuracy with more than 90% and maximum accuracy of 95.89% was obtained by Subject S3. Standard deviations between 1.32 to 1.41, which indicate low variability in performance among all the subjects. These results showed that the DWT with KNN combination can be used for real-time BCI applications and can offer a reliable communication method for people with motor impairment and disabilities.

Keywords—Brain-Computer Interface, Electroencephalogram, Discrete Wavelet Transform, k Nearest neighbour.

I. INTRODUCTION

Neurological disorders like epilepsy, amyotrophic lateral sclerosis (ALS) and cerebral palsy (CP) affect communication in many people's lives. These disorders cause various kinds of discomfort, like people with epilepsy sometimes have comprehension and language disorders, signs like paralysis of certain muscles result in complete loss of speech in people with ALS, even CP affects an individual's communication by affecting their neurological coordination, with some intellectual disabilities. Brain computer interfaces (BCIs) acts as an important improvement in the medical and scientific field, that are changing the way people can interact with the help of this technology. In healthcare, BCIs are changing rehabilitation and helping communication technologies for people with severe motor disabilities, such as those with ALS, epilepsy, CP, and spinal cord injury.

The first step of a brain-computer interface (BCI) system is to capture the electrical activity of the brain by using non-

invasive sensors such as EEG electrodes cap attached to the scalp. EEG signals provide a direct measure of cortical activity with millisecond temporal resolution [1]. These electrodes pick up the very small voltage fluctuations that occur due to neural activity and then these signals are amplified and converted into digital signals for further analysis. The data collected is initially unprocessed and usually goes through a preprocessing step in which noise and artifacts are removed; this is done by applying filters to keep only the frequency bands that are relevant to the user's planned actions. In addition to helping people with ALS, epilepsy, CP, and spinal cord injury, the BCI interprets brain signals and, in turn, translates these into actions that can control computers or other devices and give a direct way of communication. While this field is growing, it is not only for the purpose of the restoration of the lost functions but also for the enhancement of human potential to translate the mental thoughts to control commands.

In our work, we have used BCI technology to analyse EEG data that may help to improve communication for people with various neurological disorders. Our study focuses on the feature extraction from the EEG signals using the Discrete Wavelet Transform (DWT) which effectively decomposes the EEG into the frequency bands, thus improving the signal-to-noise ratio and highlighting the important features for the EEG signal classification. Finally, a k-nearest neighbour (KNN) classifier will be used to train the model to learn about basic cognitive tasks. The EEG data collected from subjects is used to determine how accurately the DWT with KNN system can distinguish between four perceived actions inside their brain for the tasks like forward, stop, left, and right.

II. LITERATURE REVIEW

Jahankhani et al. (2006) observed the use of network models for classifying EEG signals. By their study, they wanted to improve diagnosis with the help of a decision support system using machine learning techniques. After the results, they came to the conclusion that wavelet-based feature extraction technique with network models can be used for medical diagnosis and also capable of achieving high accuracy [1]. Using DWT for feature extraction and a combination of SVM, multi-layer perceptron, and KNN classifier Amin et al.

(2015) collected signals from individuals while they were at rest but with their eyes open, achieved accuracy of more than 98% compared to other models used in the study, which suggested that this approach can easily analyse EEG data can be used for further recognition of cognitive tasks [2].

To diagnose and detect individuals with epilepsy by analysing their EEG data, Adeli et al. (2003) with the help of Daubechies and harmonic wavelets, they got to know that this approach can be used for understanding changing patterns in the brains of people suffering from epilepsy [3]. Hamad et al. (2016) used DWT for feature extraction and analysed EEG to detect epilepsy, where they tested on parameters like entropy, variance, skewness and relative wave energy to analyse the performance of this approach. Using this approach, they learned that it can be very useful for detecting epilepsy [4].

To analyse wavelet-based EEG data processing Faust et al. (2015) reviewed the past ten years of research studies, where they observed that approach given by Adeli was the best among the others, for developing an automatic epilepsy detection and diagnosis system with the help of EEG signals, which was more consistent, accurate and reliable [5]. Using DWT and Generalized Fractal Dimensions (GFD) Easwaramoorthy et al. (2010) for detecting epileptic activities, where they noted that there was a variable difference seen in the GFD levels of epileptic people and healthy individuals, further they also observed that multifractal analysis and wavelet decomposition was more effective in diagnosing epilepsy [6].

For diagnosing people with epileptic seizures Kumar et al. (2014) used DWT and Approximation Entropy (ApEn) of EEG signals, where the ApEn values were quantified with the help of DWT to analyse the EEG data obtained from healthy and epileptic patients, further, they also used ANN for distinguishing between seizures and non-seizures where it gave 100% accuracy and also confirmed that ApEn values for epileptic people was lower as compared to healthy individuals [7]. For classifying left-right hand motions Bhattacharyya et al. (2010) used algorithms like LDA, QDA, and KNN on BCI Competition II 2003 dataset, where they found out that KNN emerged as the best algorithm with average band power (84.29%), whereas QDA had the best wavelet coefficients (80%) and they also noted that feature vectors can be helpful in categorising movements [8].

To categorize human emotions Murugappan et al. (2011) used wavelet transform and KNN, where they collected EEG data from 20 volunteers using audio-visual stimuli and 62 electrodes, it was observed that accuracy for 62 channels was 82.87% and for 24 channels it was 78.57% and it also showed the possibility using classifiers and feature extraction techniques for identifying emotions [9]. Yean et al. (2018) used different distance functions and KNN classifier to analyse the emotional state of stroke patients and used DFA for feature extraction from EEG signals, where they showed how the effectiveness of this approach is dependent on distance functions used in this approach [10].

Hu et al. (2016) highlighted how distance functions play an important role in the performance of KNN classifier for classification of medical data, where Chi-square distance function achieved highest accuracy for all type of dataset on the other hand cosine and Euclidean distance performed poorly for mixed datasets [11]. To distinguish between the brain activity of a normal brain and an epileptic brain

Chaovalitwongse et al. (2007), observed 10 patient's EEG data on which they used time series analysis, chaos theory and KNN, where this approach gave them an average sensitivity of 81.29% and an average specificity of 72.86%, and it also confirmed that this approach can help in diagnosis of epilepsy by identifying hidden patterns in EEG data [13].

Mehmood et al. (2015) used KNN and SVM classifier for categorizing emotions, where they observed that both the classifiers can easily identify emotions based on EEG data which can be useful for emotional computing applications [15]. Using SVM and KNN classifier for classifying EEG signals, Sha Abani et al. (2020), where they observed both KNN and SVM are efficient in identifying different EEG features and patterns and also showed the advantages and disadvantages with the effectiveness of classifiers in identifying the patterns [16].

For the detection of epilepsy Choubey et al. (2021) came up with an approach where they used KNN and ANN classifiers for analysing EEG signals, based on the performance metrics they confirmed that this approach gave accurate results for identifying epilepsy with fewer features required for classification, which informed about the effectiveness of this approach [17]. To distinguish between guilty and innocent subjects Bablani et al. (2018) used KNN classifier to classify the EEG signals, where they used binary classification and cross-validation, for all the features, certain performance parameters like accuracy, sensitivity, and specificity were used where mobility was one feature which gave the highest classification accuracy of 96.7%, that confirmed that this approach can be used for deception detection [18]. Survey determined that designing BCI using DWN with KNN was possible, even though some of the issues related to data acquisition methods and pre-processing. So in this study we improved these methods to check the possibilities for designing the BCI.

III. METHODOLOGY

A. Data Collection

To maintain consistency in the data, signals were collected from male subjects falling under the age group between 18 and 21, all the following subjects were right-handed. All the selected subjects were mentally stable and were happy and in good mental state without any kind of headache, disorders, and diseases. To avoid any kind of uncertainty or to avoid any kind of error during the trials, preliminary training was given to all the five subjects where they were asked to practice their tasks before the trials, they were taught to perform certain actions, and provided them the knowledge about experimental instrument like Gold plated electrodes and our BCI systems.

Then, they were asked to spell words mentally for the tasks such as FORWARD, STOP, LEFT, and RIGHT, without vocalizing. Ten trials were conducted for each task, where 12 features were extracted from each trial and two seconds were given for each trial. We took 200 samples for classification with the help of an analogue digital instrument using LabChart, sampling at 100 Hz. To identify different brain wave patterns associated with the tasks, six frequency bands were separated from each sample. Notch filtering at 50 Hz was used at the preprocessing step to eliminate the background noise and to produce clean and accurate signals for the data analysis during data acquisition.

B. Feature Extraction

Generally, wavelet transform (WT) is used to analyze the EEG signals for extracting the temporal and spectral features from the data, here we have used Discrete Wavelet Transform (DWT) which is the most common technique used for feature extraction. DWT is widely used for the time–frequency analysis of biomedical signals [2]. It is a signal processing tool used in machine learning for feature selection especially for non-stationary signals. In DWT, the inner product of the original signal with the basis wavelet function is taken at discrete points (usually dyadic to ensure orthogonality) and the result is a weighted sum of a series of basis functions [3].

DWT enables the determination of frequency bands that are most relevant for a given cognitive or motor task and which aids in the identification of relevant features from the EEG signal. The DWT decomposes a specific signal into approximation and detail coefficients at the first level. Then the approximation coefficients are additionally decomposed into the next level of approximation and detail coefficients [4][5]. It reviews EEG signals in the delta, theta, alpha, beta, and gamma frequency bands, which helps in understanding the nature of the various mental states.

DWT focuses on important coefficients of the EEG data, which reduces the high dimensionality of the data, which helps in reducing the noise and reducing the dimensionality of the data. In the context of EEG-based motor imagery classification, DWT also uses entropy-based features to improve the accuracy of the classification.

DWT and its parameters are defined with the help of given formula in Eq.1 [6][7].

$$DWT(g, h) = \frac{1}{\sqrt{|2^g|}} \int_{-\infty}^{\infty} x(t) \psi\left(\frac{t - 2^g h}{2^g}\right) dt \quad (1)$$

Here g is the scale parameter that defines that how much a wavelet is stretched and compressed, h defines the translation parameter that defines change in wavelet function $\psi(t)$ along the time axis, $x(t)$ is the input signal, $(t - 2^g h)/2^g$ is the translated and scaled version of the function and the integration function is used to compute the correlation between signal and the wavelet.

C. Classifier

The k nearest neighbours (KNN) method is an approach in machine learning that is generally used for classification and regression tasks because of its simplicity and efficiency. It can be used for examining EEG signals for brain mapping and neurosemantics purposes. The performance of this algorithm greatly depends on two factors: a suitable similarity function and an appropriate value for k . If k is too large, big classes will overwhelm the small classes, whereas if k is too small, the advantage of KNN algorithm is not exhibited [8]. This classifier makes a decision on comparing a new sample (testing and unseen data) with the baseline data (training data) [9]. The algorithm focuses on data points that are close to each other and tend to have category labels. While classifying a point, it looks at the majority class, among its k neighbours in the feature space. Initially, the system measures the gap between the added data point and all previous training data points by utilizing methods, like distance calculation. Since

KNN is a distance-based algorithm, the classification performance is affected by the calculation of the distance [10][11]. To calculate the distance, we use the Euclidean distance formula [12] as shown in Eq.2.

$$E.D. = \sqrt{(u_2 - u_1)^2 + (v_2 - v_1)^2} \quad (2)$$

Here, data points are represented by u and v coordinates the process involves identifying the k data points and deciding which class label appears most frequently among them. The ideal value for k is typically established through cross validation to maintain a balance between model complexity and overall performance.

To evaluate the performance of k-NN we use confusion matrix, we consider four subsets of classification results: 1) true positives (TPs), which denote correct classifications of positive cases; 2) true negatives (TNs), which denote correct classifications of negative cases; 3) false positives (FPs), which denote incorrect classifications of negative cases into class positive; and 4) false negatives (FNs), which denote incorrect classifications of positive cases into class negative [13]. This matrix gives Accuracy, Precision, Recall (Sensitivity), and the F1 Score, each one of them specifying valuable insights with respect to the performance of k-NN classifier.

Accuracy measures the proportion of correctly predicted instances out of the total instances as given in Eq.3 [14][15]:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

Precision shows the accuracy of positive predictions by calculating the correctly predicted positive observations with respect to the total predicted positive outcome as shown in Eq.4 [14][15]:

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

Recall (Sensitivity) tells us about the classifier's ability to identify actual positive instances, computed as the r correctly predicted positive observations with respect to all actual positives as shown in Eq.5[14][15]:

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

F1 Score indicates the performance of the classifier by considering both Precision and Recall. It is the harmonic mean of Precision and Recall as given in Eq.6[14][15]:

$$F1 \text{ score} = \frac{Precision \cdot Recall}{Precision + Recall} \quad (6)$$

These metrics play a crucial role in understanding the behaviour of the classifier across various scenarios, such as distinguishing between different mental states or identifying a particular pattern in BCI systems for analysing the EEG signals.

IV. RESULT AND DISCUSSION

We evaluated the performance of the study by implementing DWT features with KNN classifier, for this study, totally five subjects participated from the age group between 18 and 21, to study the various patterns generated by different subjects and to determine consistent results for all subjects. Fig.1 indicated that all the subjects achieved mean accuracy more than 90%, where the highest mean accuracy of 93.90% was achieved by S3 and the lowest mean accuracy of 92.18% by S4. Fig.2 shows that even the best maximum accuracy achieved was 95.89% for S3 and the minimum accuracy was 94.92% by S4, where the average maximum accuracy of 95.31% and the average minimum accuracy of 90.28% was achieved. From Table.1 we also get to know that S3 recorded the least standard deviation of 1.32 and the highest standard deviation of 1.41 was recorded by S4.

Table.1 indicates that the least training time of 64.01 seconds was taken by S3 and the highest training time of 65.10 seconds was taken by S2. Based on these observations, we get to know that S3 performed the best among all the subjects, which might be because of better environmental factors during the trials, and sometimes minimal errors in the instruments recording the signals can affect the results. Considering the training time, mean accuracy, and standard deviation, the combination of DWT with KNN gave overall good results for all the subjects.

TABLE I. CLASSIFICATION ACCURACY USING DWT FEATURES WITH KNN CLASSIFIER

S N o	Subj ect	Traini ng Time in sec	Testin g Time in sec	Performance Accuracy			
				Max	Min	Mean	SD
1	S1	64.78	0.96	95.67	90.16	92.89	1.39
2	S2	65.10	0.99	95.11	90.54	93.23	1.34
3	S3	64.01	0.94	95.89	90.90	93.90	1.32
4	S4	64.89	0.98	94.92	89.84	92.18	1.41
5	S5	64.90	0.98	94.98	89.96	92.64	1.40

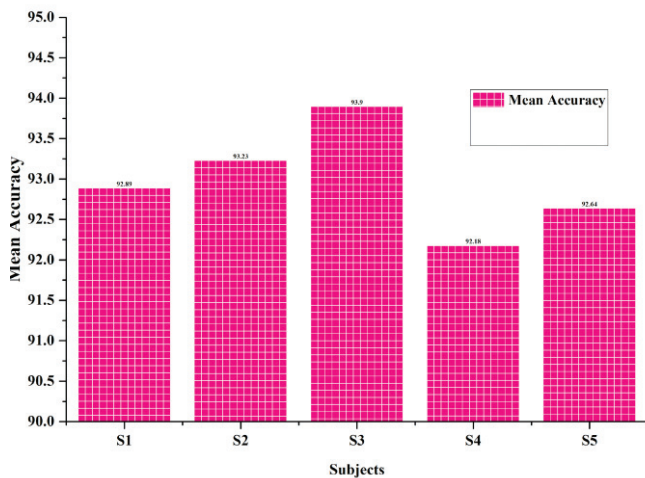


Fig. 1. Mean Accuracy

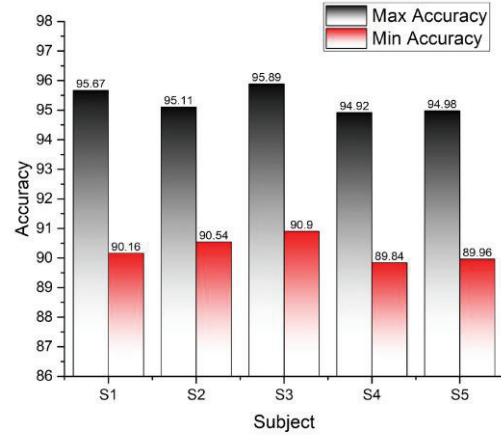


Fig. 2. Maximum and Minimum Accuracy

V. CONCLUSION

The experiment was conducted with 5 subjects, by placing gold-plated electrodes on their skull and analyzing the EEG signals generated from their brain with the help of BCI, where they were requested to imagine four tasks like stop, forward, left, and right mentally. Features were extracted using DWT and a KNN classifier was used to train them further. After analysing all the trials and experiment results, we got to know that training time for all subjects was less than one second each and training times were always between 64 and 65 seconds, suggesting that the system could be applicable for real time use even overall performance was also steady with standard deviations of 1.32 to 1.41. This low variability shows that the model is able to interpret consistently the EEG data from one trial to another while showing high accuracy rates.

The result of the study shows that K-nearest neighbour classifier combined with Discrete Wavelet Transform for feature extraction is suitable for developing real-time brain computer interface applications in improving the quality of life of people with neurological conditions such as cerebral palsy, epilepsy, and ALS where communication may be severely compromised.

VI. FUTURE STUDY

We planned to conduct our study in online mode to recognize certain tasks in real time by using individual trial analysis to improve the performances of the system.

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